Supplementary Material: Learning-Aware Lyapunov Resource Control for Correlated Multi-Agent Markov Learning

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I. Proof of the Same-Stream Actual-Risk Enclosure

Consider the stationary common factor in the main paper, with independent private observation noise and a deterministic probe $d = (q_d, b_d, n_d)$ ending at $T = n_d b_d$. The probe sigma-field $\mathcal{H}_d$ contains its observations; spatial contrasts contain no additional information about the common factor. Under true instance j, Gaussian filtering gives

$$C_T \mid \mathcal{H}_d \sim \mathcal{N}(m_j, P_j), \quad m_j \sim \mathcal{N}(0, s_j^2), \quad s_j^2 = \theta_j - P_j. \tag{S1}$$

The posterior variance P_j is deterministic because the probe schedule is fixed before data. Starting from variance θ_j, the observation update is $P \leftarrow P/(1+q_d P)$; between observations it is $P \leftarrow \lambda^{2b_d} P + (1-\lambda^{2b_d})\theta_j$. The last update has no subsequent prediction.

Fix either downstream program a, including its budget, initialization, sample spacing and deterministic weights. All learning observations occur strictly after T. With $r_a = \lambda^{b_a}$ and the notation of main Eq. (38), its unconditional stationary risk is

$$J_{j,a} = d_a^2 + \theta_j \sum_{t,u=1}^{N_a} \omega_{a,t}\omega_{a,u} r_a^{|t-u|} + q_a^{-1} \sum_{t=1}^{N_a} \omega_{a,t}^2. \tag{S2}$$

Conditional means of future common states are $\lambda^{b_a t} m_j$. Their conditional covariance is their stationary covariance minus $s_j^2 \lambda^{b_a(t+u)}$. Private learning noise is independent of the probe. Therefore

$$\mathbb{E}_j[Z_a^2 \mid \mathcal{H}_d] = J_{j,a} + 2d_a k_a m_j + k_a^2(m_j^2 - s_j^2). \tag{S3}$$

This identity includes the posterior mean, which is unbounded across probe histories; no uniform conditional Gaussian bound is invoked.

Let E_j be the event that the decision selects the wrong frozen program, and let $p_j = \Pr_j(E_j) \le \delta$. Subtracting Eq. (S3) for the correct and wrong programs gives

$$\mathbb{E}_j Z_{\widehat{a}}^2 = J_{j,a_c} + p_j \Delta J_j + \mathbb{E}_j[\mathbf{1}_{E_j} U_j], \tag{S4}$$

where

$$U_j = 2(d_{a_w} k_{a_w} - d_{a_c} k_{a_c}) m_j + (k_{a_w}^2 - k_{a_c}^2)(m_j^2 - s_j^2). \tag{S5}$$

The centered Gaussian first and second moments are orthogonal, with variances s_j^2 and $2s_j^4$. Consequently $\mathbb{E}_j U_j = 0$ and $\mathbb{E}_j U_j^2 = \Gamma_j^2$ as defined in the main paper. Cauchy–Schwarz gives $|\mathbb{E}_j[\mathbf{1}_{E_j} U_j]| \le \sqrt{p_j}\Gamma_j \le \sqrt{\delta}\Gamma_j$. Bounding the signed term $p_j \Delta J_j$ and using nonnegative squared risk proves main Theorem 5.

II. Exact Weights of the Implemented Learners

Let $M_a = \min\{\lfloor B'_m/(h+q_a)\rfloor, \lfloor B'_e/b_a\rfloor\}$ be the number of scheduled observations after subtracting probe costs. All M_a observations are paid. The FIFO withholding convention leaves $N_a = (M_a - D)_+$ usable observations, not a free replacement budget. For $N_a = 0$ the reported parameter is one, so $d_a = 1$ and the weight vector is empty.

A. Running mean

The Layer-A output is $(1+\sum_{t=1}^{N_a} \bar{X}_t)/(N_a+1)$. Thus $d_a = 1/(N_a+1)$ and every $\omega_{a,t} = 1/(N_a+1)$. Main Eq. (38) then gives

$$k_a = \frac{r_a(1 - r_a^{N_a})}{(N_a+1)(1-r_a)}. \tag{S6}$$

B. Averaged FIFO affine learner

In Layer B, an increment $\eta(\bar{X}_t - \mu w_{t-1})$ is computed before enqueueing and is applied after D opportunities. The original implementation fixes $\mu = 1/2$ and $\eta = 0.1/[\mu(D+1)]$, starts from parameter one with an empty queue, and averages the applied parameters. Indexing these applied parameters as z_n gives

$$z_n = z_{n-1} - \eta\mu z_{n-D-1} + \eta \bar{X}_n, \quad z_{-D} = \cdots = z_0 = 1. \tag{S7}$$

Define $h_0 = 1$, $h_t = 0$ for $t < 0$, and $h_t = h_{t-1} - \eta\mu h_{t-D-1}$ for $t \geq 1$. For $N = N_a > 0$, the output $N^{-1}\sum_{n=1}^{N} z_n$ has exact coefficients

$$\omega_{a,t} = \frac{\eta}{N}\sum_{v=0}^{N-t} h_v, \quad d_a = 1 - \mu\sum_{t=1}^{N}\omega_{a,t}. \tag{S8}$$

To verify the deterministic term, constant input $\bar{X}_n = \mu$ keeps the initial history at one; linearity then yields the identity for d_a. Eq. (S8) represents the average of applied iterates; all scheduled observations remain charged.

III. Shrinking-Risk and Budget-Shift Interface

Suppose the finite action catalogue has $|\lambda| \leq 1 - \gamma < 1$, bounded D, and strictly stable affine recursions. For the registered step rule this stability follows directly from the characteristic polynomial $z^{D+1} - z^D + t$, $t = \eta\mu$. At small $t > 0$, the root at one moves inside the unit disk and the other roots remain inside. Writing a first unit-circle crossing as $z = e^{i\omega}$ yields $t = 2\sin(\omega/2)$ and the smallest positive crossing $\omega = \pi/(2D+1)$. Thus $0 < t < 2\sin[\pi/(4D+2)]$ is stable; the registered $t = 0.1/(D+1)$ lies strictly inside this interval. The impulse response consequently decays geometrically, and $\sum_{v\geq 0} h_v = 1/(\eta\mu)$.

The interior weights in Eq. (S8) are $1/(\mu N)$ up to a summable end correction: their difference from these constant weights has ℓ_1 norm $O(N^{-1})$ and maximum absolute value $O(N^{-1})$. The homogeneous transient is summable, so its average d_a is $O(N^{-1})$. The AR(1) covariance kernel has uniformly bounded absolute row sums. Substitution into Eq. (S2) therefore gives

$$J_{j,a} = \frac{1}{\mu^2 N_a}\left[\frac{1}{q_a} + \theta_j\frac{1+r_a}{1-r_a}\right] + O(N_a^{-2}). \tag{S9}$$

The running mean has the same expansion with $\mu = 1$. These constants are uniform over the finite catalogue and bounded instance class. On a budget ray, $N_a = s/c_a + O(v+D+1)$ with $c_a = \max\{(h+q_a)/\beta_m, b_a/\beta_e\}$ and fixed probe charge v. Hence $J_{j,a} = K_{j,a}/s + O((v+D+1)/s^2)$, with

$$K_{j,a} = \frac{c_a}{\mu^2}\left[\frac{1}{q_a} + \theta_j\frac{1+r_a}{1-r_a}\right]. \tag{S10}$$

Subtracting a fixed probe charge changes N_a by at most $O(v+D+1)$, giving the claimed budget-shift remainder. Moreover $k_a = O(N_a^{-1})$ because the weights are uniformly $O(N_a^{-1})$ and $\sum_{t\geq 1}|r_a|^t$ is bounded. Since $s_j^2 \leq \theta_j$, main Eq. (39) yields

$$\sqrt{\delta}\Gamma_j = O(\sqrt{\delta}/s^2). \tag{S11}$$

For a pair of frozen actions satisfying the main phase theorem's leading-gap and wrong-decision assumptions, adding this remainder to admission changes only its second-order class constants. The same necessary-information and sufficient-scale argument then applies to the corrected actual-risk rule. This is an averaged, additive Gaussian affine result; it does not remove the nonzero noise floor of a fixed-step unaveraged learner or the conditional-history requirements of the general Markov theorem.

TABLE I
Population enclosures for mean information-only minus learning-aware risk, rounded outward. All means use the listed complete stratum.

Layer	Stratum	Cells	Lower	Upper
A	All	1184	.0506	.0706
A	Intermediate	576	.0770	.1180
A	Original practical	446	.1607	.1952
A	Search-censored	32	.4879	.4880
B	All	608	.0079	.0223
B	Intermediate	288	-.0049	.0252
B	Original practical	222	.0562	.0773
B	Search-censored	32	.1957	.1958

IV. Complete Existing-Evidence Reanalysis

The calculation uses all registered EXP-016B budget points, both regimes and all eight policies: 14,336 policy cells, each with its original 192-seed empirical mean. Layer A contains 104 scenario families and Layer B its original 56, including eight search-censored families in each layer. For fixed policies Eq. (S2) is exact. For probe policies main Eq. (40) supplies population-risk bounds. When information-only and learning-aware policies execute the same frozen program, their paired difference is identically zero, not the difference of two independent interval bounds. Otherwise subtract their one-sided bounds.

Below identification, all 208 Layer-A and 112 Layer-B comparisons are identical programs; above finite value thresholds the corresponding counts are 384 and 192. In the original practical subset, 323/446 Layer-A and 176/222 Layer-B cells have strictly positive improvement lower bounds, while 101 and 33 have strictly negative upper bounds; the remaining 22 and 13 are inconclusive. These include regime-specific losses even when the stratum-average improvement is positive. No new admission rule was executed and no old gate was replaced. The old probe-length routine used the sufficient exponent $\log(0.5/0.025)$, giving an equal-prior average-error target of 0.025 rather than a directional guarantee of 0.025. The reanalysis uses each probe's exact affinity $\exp(-I_B)$ as a valid bound on either directional error; over all 104 frozen probe designs these bounds range from 0.0352271 to 0.0497465. Failure of this sufficient directional certificate does not establish that the actual

directional errors exceed 0.025. No outcomes or probe lengths are changed to obtain a smaller error bound. This outcome-known reanalysis gives population enclosures, not new confidence intervals for empirical means.

V. Complete Threshold-Sandwich Scale Calculation

For the independently initialized Gaussian running mean, let $c_a = \max\{(h+q)/\beta_m, b/\beta_e\}$ and $r = \lambda^b$. The exact horizon is $N_a = s/c_a + O(1)$, with bounded terminal charges and rounding. Main Eq. (8) and Eq. (13) give $K_j(a) = c_a\{q^{-1} + \theta_j(1+r)/(1-r)\}$ and a uniform $O(s^{-2})$ remainder. Removing probe charge v changes N_a by $O(v+1)$ and hence risk by $O((v+1)/s^2)$ for $s \geq s_0(v+1)$; compact persistence and the finite catalogue make these constants uniform. Thus the phase-theorem risk conditions have a concrete Gaussian realization, separately from the constant-step affine certificate. Write $d_\delta = \mathrm{kl}(1-\delta, \delta)$, $r_0 = 1-\gamma < 1$, and $Q = \max \mathcal{Q}$. Under either true instance, predictable spacing gives $\|C_{S_k}\|_{L^2} \leq r_0 \|C_{S_{k-1}}\|_{L^2} + \sqrt{\theta_+}$, hence a uniform bound $M_C = \sqrt{\theta_+}/(1-r_0)$. The normalized observation $Y_k/\sqrt{q_k}$ has L^2 norm at most $M_C + 1$. For either candidate filter, write its posterior mean as $(1-t_{j,k})m^-_{j,k} + t_{j,k}Y_k/\sqrt{q_k}$, with predictable $0 \leq t_{j,k} \leq 1$. Prediction multiplies this mean by a factor of magnitude at most r_0; the triangle inequality bounds both predictive means uniformly in L^2 by $M_F = r_0(M_C+1)/(1-r_0)$, starting from zero. Both predictive variances are at most θ_+, so $1 \leq V_{j,k} \leq V_+ = 1 + Q\theta_+$. Main Eq. (26) gives the expected, rather than pointwise, bound $\mathbb{E}_j \mathcal{I}_{j,1-j}(Z_k, A_k) \leq i_+$, where $i_+ = (\log V_+ + V_+ + 4QM_F^2)/2$ is finite. Extend a stopped rule predictably after stopping; nonnegativity and its deterministic budget bound $n_{\max}(s) \leq c_B s$ imply $\mathbb{E}_j \sum_{k \leq \tau} \mathcal{I}_{j,1-j} \leq i_+ n_{\max}(s)$. Together with main Eq. (30), this proves $B_N \geq c_N d_\delta$. The fixed probe has information at least $i_- > 0$, so $n_\delta = \lceil i_-^{-1} \log(1/\delta) \rceil$ observations suffice and cost at most $c_p n_\delta + D_{\max}$. Continuity of $\log(1/\delta)/d_\delta$ gives $n_\delta \leq c_0 + c_1 d_\delta \leq c_0 + c_2 B_N$. Write $u_\delta \leq c_p n_\delta + D_{\max}$ for the full probe charge. The assumed expansion and budget-shift inequality give a high-instance improvement over the full-budget fallback certificate of at least $g_{\min}/s - \{2M + L(u_\delta+1)\}/s^2 - \delta w_+/s$. It is strictly positive for $s > \{4M + 2L(u_\delta+1)\}/g_{\min}$, also requiring $s \geq s_0(u_\delta+1)$. On instance zero, the assumed coefficient ordering gives $\epsilon_{\mathrm{safe}} \leq \{2M + L(u_\delta+1)\}/s^2 + \delta_0 w_+/s$. Thus $s \geq \sqrt{2\{2M + L(u_\delta+1)\}/\bar{\epsilon}} + 2\delta_0 w_+/\bar{\epsilon}$ suffices. Using $2\sqrt{xy} \leq x+y$, all three requirements are bounded by a class-constant multiple of $n_\delta + B_{\mathrm{oracle}} + B_\epsilon + 1$. Taking the maximum proves the upper bound in main Eq. (41); its lower bound follows because every admitted δ-correct probe obeys both information constraints.

VI. Robust-Certificate Sensitivity Details

This section supplies the proof of main Corollary 2. Write $\delta^+ = C_q(r_q^+)^b$. On the coverage event, $\delta^+ \geq \delta$ and monotonicity on $\mu - 2L\delta^+ > 0$ gives $a^+ \geq a$ and $\beta^+ \geq \beta$. Evaluating the delayed Young inequality at the shared $\lambda^+ = \sqrt{h/a^+}$ proves the upper recursion; separately optimized forcing terms need not be monotone. For $\Delta\delta = \delta^+ - \delta$, $\Delta K = K_q^+ - K_q$, and $\Delta\Omega = \Omega_q^+ - \Omega_q$,

$$\begin{aligned}
\Delta a &= 2\eta^2 \Delta K + (2\eta L + 4\eta^2 L^2)\Delta\delta, \\
\Delta\beta &= 2\eta^2 \Delta\Omega + 4\eta^2 G^2 \Delta\delta \\
&\quad + 4\eta G^2 \left[\frac{(\delta^+)^2}{\mu - 2L\delta^+} - \frac{\delta^2}{\mu - 2L\delta} \right], \\
\Delta c &= (\sqrt{a^+} + \sqrt{h})^2 - (\sqrt{a} + \sqrt{h})^2.
\end{aligned} \tag{S12}$$

For $h > 0$,

$$\begin{aligned}
\Delta d \leq (1 + \sqrt{h/a^+})\Delta\beta + \sqrt{h}\left(\frac{1}{\sqrt{a}} - \frac{1}{\sqrt{a^+}} \right)\beta \\
+ \frac{\sqrt{a^+} - \sqrt{a}}{\sqrt{h}} g,
\end{aligned} \tag{S13}$$

and $\Delta d = \Delta\beta$ when $h = 0$. For $U(c,d) = d/(1-c) + c^n(R_0 - d/(1-c))_+$, $|\partial_d U| \leq (1-\bar{c})^{-1}$ and $|\partial_c U| \leq nR_0 + \bar{d}(1-\bar{c})^{-2}$. The mean-value bound proves main Eq. (43); applying it to the upper- and exact-score minimizers gives $U_{\widehat{a}} \leq U_{a^\star} + 2\max_a \varepsilon_a$.

VII. Stochastic-Delay Interpretation

The controlled stochastic delay differential equation associated with one fixed resource action is

$$dE(t) = -A_q E(t - \tau(t))dt + \sqrt{\eta}\, \Sigma_q^{1/2}(E_t) dW(t). \tag{S14}$$

Here $\Sigma_q(E_t)$ retains both additive innovation covariance and the state-dependent quadratic variation of random temporal-difference Jacobians. For $V(E) = \|E\|^2$, Itô's formula gives

$$\mathcal{L}_a V(E_t) = -2E(t)^\top A_q E(t - \tau(t)) + \eta \,\mathrm{tr}\, \Sigma_q(E_t). \tag{S15}$$

Participation controls spatial covariance, spacing controls mixing, and η controls their scale; the discrete theorem retains finite steps, integer delays and rounding.